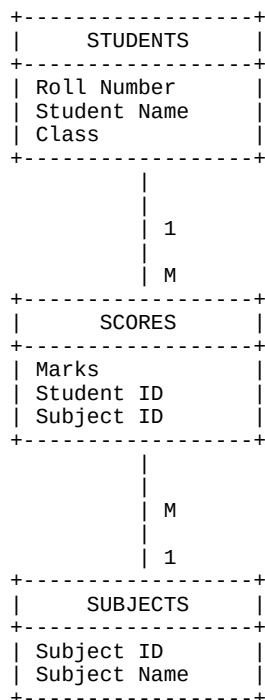


MySQL Practice Activity - School Examination System

Instructions

Using the ER Diagram and business rules, identify tables, columns, primary keys, foreign keys, data types and relationships. Create the database and solve all queries. No JOIN should be used.

School Examination System ER Diagram



Business Rules

Every student has a unique Roll Number.

Every subject has a unique Subject ID.

A student can have marks in many subjects.

A subject can be studied by many students.

Marks are stored in a separate Scores table.

One student can have only one score record per subject.

Scores table should connect Students and Subjects.

A score cannot exist without a valid student.

A score cannot exist without a valid subject.

Section A: Database Design

1. Identify all tables required.
2. Identify the Primary Key of each table.
3. Identify the Foreign Keys.
4. Create all tables.
5. Apply appropriate constraints.
6. Ensure that one student cannot have multiple score records for the same subject.

Section B: Sample Data Entry

1. Insert at least 5 students.
2. Insert at least 4 subjects.
3. Insert at least 10 score records.

Section C: Query Writing

1. Display all students.
2. Display all students of class 9A.
3. Display all students of class 10B.
4. Display student having Roll Number 3.
5. Display students sorted by name.
6. Display all subjects.
7. Display subjects sorted alphabetically.
8. Display subject having Subject ID 2.
9. Display all score records.
10. Display score records having marks greater than 80.
11. Display score records having marks less than 60.
12. Display score records having marks between 70 and 90.
13. Display score records sorted by marks ascending.
14. Display score records sorted by marks descending.
15. Display score records for Subject ID 1.
16. Display score records for Student ID 2.

Bonus Challenge (Real School Scenario)

1. Who are all the students of Class 9A?
2. Which students scored above 90 marks?
3. Show all subjects available in the school.
4. Show all score records from highest to lowest marks.
5. Find all students whose name starts with 'R'.
6. Find all students whose name ends with 'h'.
7. Display all score records where marks are between 75 and 95.
8. Display all students ordered by class and then by name.

Allowed SQL Concepts

CREATE TABLE

PRIMARY KEY

FOREIGN KEY

INSERT

SELECT

WHERE

ORDER BY

LIKE

No JOIN is required.